A SMOOTH-AT-THE-GENERIC-FIBRE MAP IS SMOOTH AT P WHEN THE CONORMAL COKERNEL OVER LAMBDA IS FREE

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Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Cokernel of the conormal map. Algebra, lemma differential-seq (cross-chapter, named). The cokernel of I/I^2 -> Omega_{B/R} tensor A is Omega_{A/R}.
- Structure of smooth algebras over a field. Algebra, lemma characterize-smooth-over-field (cross-chapter, named). Omega of a smooth algebra over a field is free of rank the relative dimension.
- Fibre dimension upper bound on opens. Algebra, lemma dimension-fibres-bounded-open-upstairs (cross-chapter, named). Gives dim_p(A/R) at least d from the specialization p of q.
- Smoothness from flatness and geometrically regular fibres. Algebra, lemma flat-fibre-smooth (cross-chapter, named). Concludes smoothness at p.

Source reference. Original reference 00QD: focused-seeds/geometry/S9-smo-neron-when-smooth/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00TT: focused-seeds/geometry/S9-smo-neron-when-smooth/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00RU: focused-seeds/geometry/S9-smo-neron-when-smooth/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00QH: focused-seeds/geometry/S9-smo-neron-when-smooth/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00TF: focused-seeds/geometry/S9-smo-neron-when-smooth/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BJ2: sources/stacks-smoothing/pinned-original-source.tex, source lines 1085–1085. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0BJ2 assume that $R \to A$ is smooth at $\mathfrak{q}$ and that we have a surjection of R-algebras $B \to A$ with kernel I. Assume $R \to B$ smooth at $\mathfrak{p}_B = (B \to A)^{-1}\mathfrak{p}$. If the cokernel of*

$$I/I^2 \otimes_A \Lambda \to \Omega_{B/R} \otimes_B \Lambda$$

is a free Λ-module, then $R \to A$ is smooth at $\mathfrak{p}$.

Proof. The cokernel of the map $I/I^2 \to \Omega_{B/R} \otimes_B A$ is $\Omega_{A/R}$, see Algebra, Lemma 00RU. Let $d = \dim_{\mathfrak{q}}(A/R)$ be the relative dimension of $R \to A$ at $\mathfrak{q}$, i.e., the dimension of $\mathrm{Spec}(A[1/\pi])$ at $\mathfrak{q}$. See Algebra, Definition 00QD. Then $\Omega_{A/R,\mathfrak{q}}$ is free over $A_{\mathfrak{q}}$ of rank d (Algebra, Lemma 00TT). Thus if the hypothesis of the lemma holds, then $\Omega_{A/R} \otimes_A \Lambda$ is free of rank d. It follows that $\Omega_{A/R} \otimes_A \kappa(\mathfrak{p})$ has dimension d (as it is true upon tensoring with $\Lambda/\pi\Lambda$). Since $R \to A$ is flat and since $\mathfrak{p}$ is a specialization of $\mathfrak{q}$, we see that $\dim_{\mathfrak{p}}(A/R) \geq d$ by Algebra, Lemma 00QH. Then it follows that $R \to A$ is smooth at $\mathfrak{p}$ by Algebra, Lemmas 00TF and 00TT. $\square$